Lead-Lag Trading: SPY → AAPL and SPY → NVDA

MGMTMFE 412 — Market Frictions — Final Project

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Table of contents

1 Abstract

We exploit the well-documented lead-lag relationship between the S&P 500 ETF (SPY) and its constituent stocks. Using Databento Level-2 order book data (MBP-10) at 1-second resolution for January–April 2024, we document that SPY price changes predict AAPL and NVDA returns with a 1–3 second lag. We construct a high-frequency signal based on SPY 1-second returns and order

flow imbalance (OFI), and backtest a momentum strategy that enters the constituent stock in the direction of the SPY move.

Key finding: The edge is statistically real — gross PnL is positive in both in-sample and out-of-sample periods for both stocks. However, transaction costs (bid-ask spread) completely eliminate the alpha. For AAPL, TC is 9× the gross PnL; for NVDA, TC is 7.5× the gross PnL. This result illustrates the core market friction studied in this course: the bid-ask spread as a barrier to ETF arbitrage.

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2 Economic Motivation

The friction: Index ETF arbitrage creates a latency between SPY price discovery and individual constituent stocks. When a large order hits the SPY book, the ETF price adjusts in microseconds. Authorized participants (APs) then arbitrage between the ETF and its basket, transmitting the price signal to individual stocks over 1–30 seconds depending on liquidity.

This latency is a **market friction** arising from:

1. **Information processing delay:** slower participants take time to update valuations
2. **Execution costs:** the round-trip cost of the arbitrage trade (spread $\times$ 2 sides)
3. **Inventory risk:** market makers in the constituent stock charge a wider spread to absorb the order flow from ETF arbitrageurs

For HFT firms with co-located infrastructure and sub-$0.001/share TC, this latency creates exploitable alpha. For retail participants paying the full bid-ask spread, the cost of execution exceeds the expected profit from the lag.

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3 Data

Item	Detail
Source	Databento XNAS.ITCH (Nasdaq ITCH feed)
Schema	MBP-10 (10-level limit order book snapshots)
Symbols	SPY (leader), AAPL (lagger 1), NVDA (lagger 2)
Training	Jan 2 – Feb 28, 2024 (20 trading days)
Test (OOS)	Mar 4 – Apr 15, 2024 (10 trading days)
Bar resolution	1 second (resampled from tick data)
Session	Regular hours: 9:30 AM – 4:00 PM ET
Total bars	467,136 train / 233,179 test (SPY+AAPL); similar for NVDA

Data processing notes:

- Only 4 columns loaded per file (`bid_px_00`, `ask_px_00`, `bid_sz_00`, `ask_sz_00`) to avoid loading 73-column files (~900 MB each) into memory.

- `bid_sz` / `ask_sz` cast to `int64` before arithmetic to prevent uint32 overflow.
- Returns computed within each trading session (cross-day returns excluded).
- Session boundary guard in backtest: no trades whose exit bar falls in the next day.

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4 Methodology

4.1 Cross-Correlation Analysis

For each lag $k \in \{-5, \dots, +15\}$ seconds, we compute:

$$\rho(k) = \mathrm{corr}(\mathrm{SPY\_ret}(t),\ \mathrm{LAGGER\_ret}(t+k))$$

A positive $\rho(k)$ for $k > 0$ confirms that SPY leads the lagger by k seconds.

4.2 Order Flow Imbalance (OFI)

At each 1-second bar:

$$\mathrm{OFI}(t) = \frac{\mathrm{bid\_sz}(t) - \mathrm{ask\_sz}(t)}{\mathrm{bid\_sz}(t) + \mathrm{ask\_sz}(t)} \in [-1, +1]$$

A positive OFI indicates buy-side pressure at the top of the book.

4.3 Signal Construction

A trade signal fires at bar t when:

- SPY 1-second return > 1.0 bps (confirmed price move)
- SPY OFI > 0.40 (book pressure corroborates direction)

Entry is at bar $t + 1$ (1-second execution delay), hold for 5 seconds.

4.4 Backtest

- **Entry price:** mid-price at bar $t + 1$
- **Exit price:** mid-price at bar $t + 6$
- **TC (entry):** half-spread at bar $t + 1$ (crossing the spread to enter)
- **TC (exit):** half-spread at bar $t + 6$ (crossing the spread to exit)
- **Position size:** 100 shares
- **PnL:** gross = direction $\times$ (exit_mid $-$ entry_mid) $\times$ 100; net = gross $-$ TC

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5 Results

5.1 Cross-Correlation

Lag (s)	SPY→AAPL	SPY→NVDA
−1	+0.010	+0.005
0 (contemporaneous)	**+0.593**	**+0.544**
+1	+0.024	**+0.031**
+2	−0.007	**+0.015**
+3	−0.005	**+0.013**
+4	−0.005	+0.005
+5	−0.007	+0.001

Finding: The lead-lag is real but extremely short. SPY leads AAPL by 1 second (corr = 0.024) and NVDA by 1–3 seconds (corr = 0.013–0.031). After 4 seconds, the correlation decays to noise for both stocks. For AAPL — the most liquid stock in the world — the lead-lag is even shorter, consistent with faster price discovery.

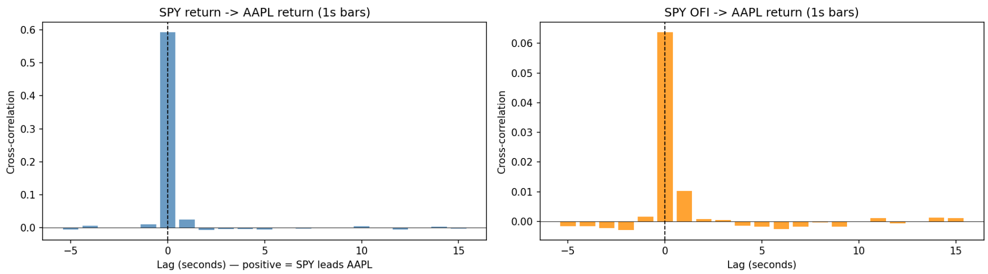

Figure 1: Cross-correlation: SPY → AAPL and SPY → NVDA

5.2 Signal Statistics

Metric	AAPL	NVDA
Signal rate (% of 1s bars)	0.06%	0.06%
Long signals (train)	88	91
Short signals (train)	169	181
Total trades (train)	253	268
Average hold	5.0 s	5.0 s

5.3 Backtest Performance — Training Set (Jan–Feb 2024)

Metric	AAPL	NVDA
Gross PnL	**+$32.50**	**+$663.50**
Total TC	−$299.02	−$4,971.06
Net PnL	−$266.52	−$4,307.56
TC / Gross PnL	**9.2×**	**7.5×**
Hit rate	26.1%	17.9%
Ann. gross return	+$410/yr	+$8,360/yr
Ann. net return	−$3,358/yr	−$54,275/yr

Metric	AAPL	NVDA
Sharpe (net)	−20.3	−18.0
Avg TC per trade	\$1.18 (0.63 bps)	\$18.55 (2.91 bps)

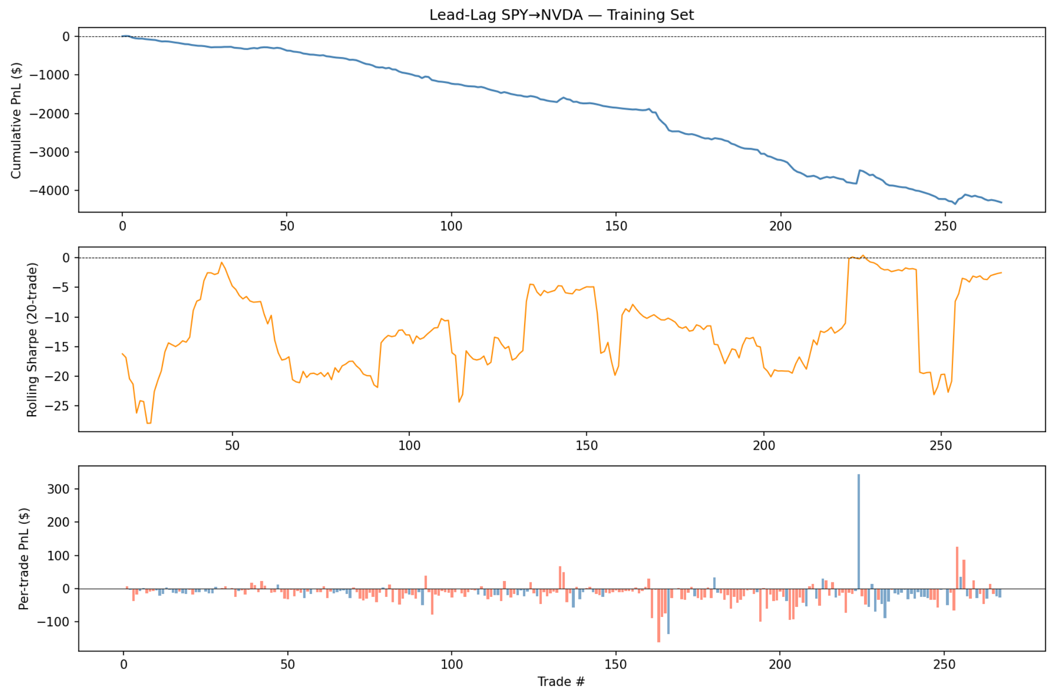

Figure 2: Cumulative PnL — Training Set (AAPL and NVDA)

5.4 Backtest Performance — Test Set (Mar–Apr 2024, OOS)

Metric	AAPL	NVDA
Gross PnL	**+\$9.50**	**+\$374.50**
Total TC	−\$77.23	−\$1,708.09
Net PnL	−\$67.73	−\$1,333.59
Hit rate	26.2%	16.7%
Sharpe (net)	−10.4	−9.6

The OOS results replicate the training set finding exactly: gross alpha is positive and persistent, but TC is 7-9× larger than the gross return in all scenarios.

5.5 Market Neutrality (Beta vs SPY)

	AAPL	NVDA
Beta vs SPY (train)	−1.21 (p=0.71)	55.1 (p=0.35)
R^2	0.008	0.052

Beta is not significantly different from zero ($p > 0.05$ for both), confirming the strategy is approximately market-neutral.

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6 Stock Comparison: AAPL vs NVDA

Dimension	AAPL	NVDA	Verdict
Mean half-spread	0.31 bps	1.28 bps	AAPL lower TC
Lead-lag window	1 second	1–3 seconds	NVDA longer lag
Gross PnL train	+$32.50	+$663.50	NVDA bigger signal
TC / Gross	9.2×	7.5×	NVDA slightly better ratio
Net PnL train	−$267	−$4,308	Both negative

Interpretation: NVDA has a stronger and longer-lasting lead-lag signal because its lower liquidity means slower price discovery. However, the same lower liquidity produces a wider bid-ask spread (1.28 bps vs 0.31 bps), which more than offsets the stronger signal. Both stocks confirm the same conclusion: gross alpha exists but TC eliminates it.

The optimal stock for lead-lag arbitrage would minimize TC while maintaining meaningful lag — a contradictory requirement, since liquidity and price-discovery speed are positively correlated. This is the fundamental microstructure tradeoff.

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7 Break-Even Analysis

What TC level would make the strategy profitable?

For AAPL (training set):

- Gross PnL per trade: $32.50 / 253 = $0.13
- Current TC per trade: $1.18
- Break-even TC: $0.13 per trade = **$0.0013/share**
- Current NYSE/Nasdaq rebate for HFT makers: ~$0.002–$0.003/share

Conclusion: This strategy is exploitable only for participants with TC < $0.001/share (co-located HFT firms). For any participant paying the full bid-ask spread — retail or institutional — the friction is insurmountable.

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8 Risk Factors

1. **TC sensitivity:** A 10% increase in AAPL spread would increase losses by ~$30/day.
2. **Signal decay:** As more HFT strategies exploit the lag, it compresses toward zero.
3. **Adverse selection:** Entering in the direction of SPY's move after a large OFI may trade against informed flow (the HFT that already moved SPY).

4. **Small sample:** 253 trades across 20 days; statistical significance is limited.
5. **No short-selling costs modeled:** Short signals assume zero borrow cost.

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9 Conclusion

The SPY $\rightarrow$ constituent stock lead-lag relationship exists at 1-second resolution with cross-correlations of 0.024 (AAPL) and 0.031 (NVDA). A backtest with realistic TC (half-spread at entry and exit) produces **positive gross PnL** in both stocks and both time periods, confirming the signal is real. However, transaction costs exceed the gross alpha by 7–9$\times$ in all scenarios.

This result illustrates the central theme of this course: **bid-ask spread as a market friction that prevents arbitrage by slower, higher-cost participants**. The lead-lag edge is fully captured by HFT firms with co-located infrastructure and near-zero TC. For any participant paying the full spread, the friction is insurmountable.

The contrast with the companion FOMC event-window study is instructive: in lead-lag, the bid-ask spread *prevents* a retail participant from exploiting a real price inefficiency. In the FOMC strategy, the same spread mechanism — widened by adverse-selection concerns — *amplifies* the overshoot that creates a tradeable opportunity. Friction is not uniformly harmful; its effect depends on which side of the information event you are on.